

ORIGINAL ARTICLE

Development of an Integrated Monogenic and Polygenic Risk Assessment Tool for Coronary Artery Disease and Its Application in a Community-Based Health Care Biobank

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BACKGROUND: Current genetic testing for coronary artery disease (CAD) primarily targets monogenic variants in individuals with severe hypercholesterolemia. Whether supplementing monogenic testing with polygenic risk scores for CAD and Lp(a); lipoprotein[a] levels [PRS_{Lp(a)}] improves identification of high-risk individuals in the general population remains understudied.

METHODS: A genetic probability for CAD (GenProb_{CAD}), incorporating monogenic pathogenic variants (PVs), polygenic risk scores for CAD, and PRS_{Lp(a)} was developed using Cox regression in 226 145 UK Biobank participants, validated in the remaining 226 145 UK Biobank participants, and applied to 20 477 participants of the Genomic Health Initiative, a community-based health care biobank. Predictive performance was evaluated and adjusted for clinical risk factors.

RESULTS: In the UK Biobank development cohort, PVs, polygenic risk scores for CAD and PRS_{Lp(a)} were each independently associated with CAD. In the UK Biobank validation cohort, GenProb_{CAD} outperformed PVs and the polygenic risk score in predicting CAD and reclassified risk among PV carriers. GenProb_{CAD} identified 16% of participants as high risk, 46-fold more than PV carriers (0.35%), with comparable observed CAD prevalence (15.60% versus 15.43%). Nearly 50% of CAD with high-risk GenProb_{CAD} were premature. GenProb_{CAD} also independently predicted incident CAD after adjusting for clinical risk factors. Importantly, high-risk individuals defined by GenProb_{CAD} showed consistently elevated CAD incidence across all low-density lipoprotein cholesterol strata. When UK Biobank-derived coefficients and cutoffs were applied in the Genomic Health Initiative, GenProb_{CAD} substantially outperformed the PV-only strategy, identifying 1966 high-risk participants (345 developed CAD), far exceeding the 20 PV carriers detected among those with low-density lipoprotein cholesterol ≥ 190 mg/dL (5 developed CAD). A universal genetic testing strategy using GenProb_{CAD} would hypothetically identify 20.75% of all incident CAD, 69-fold higher than current monogenic testing (0.3%). Results were generally consistent across ancestry populations, although the sample size of non-European participants was limited.

CONCLUSIONS: GenProb_{CAD}, a novel integrated genetic risk tool combining monogenic PVs, polygenic risk scores for CAD, and PRS_{Lp(a)}, improves identification of individuals at high genetic risk for CAD across diverse populations.

Key Words: coronary artery disease ■ genomics ■ humans ■ lipoprotein(a) ■ prevalence

Genetic susceptibility plays an important role in the development of coronary artery disease (CAD), either directly or indirectly through its contribution

to clinical risk factors for CAD such as hypertension, obesity, and diabetes.^{1,2} Consequently, the clinical utility of genetic risk assessment for CAD may vary with

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Nonstandard Abbreviations and Acronyms

CAD	coronary artery disease
FH	familial hypercholesterolemia
GHI	Genomic Health Initiative
GWAS	genome-wide association study
HR	hazard ratio
LDL-C	low-density lipoprotein cholesterol
LLT	lipid-lowering therapy
Lp(a)	lipoprotein(a)
PRS	polygenic risk score
PRSCAD	polygenic risk score with CAD
PRSLp(a)	polygenic risk score with Lp(a)
PV	pathogenic variant
RR	relative risk
UKB	UK Biobank

age and the presence of these conditions. The predictive performance of genetic risk factors is typically stronger at younger ages, before the onset of these conditions, whereas in older individuals, its value may be attenuated as clinical risk factors become more dominant contributors to overall risk.³

Several genetic risk factors have been associated with CAD, including rare pathogenic variants (PVs) in monogenic genes linked to familial hypercholesterolemia (FH), such as *LDLR*, *APOB*, and *PCSK9*,^{4–6} common variants associated with CAD,^{7,8} and Lp(a; lipoprotein[a]) levels.⁹ Current clinical guidelines for CAD genetic testing are limited to the identification of PVs in monogenic genes among individuals suspected of having FH.¹⁰ However, data from large-scale studies indicate that this approach identifies fewer than 2% of individuals with severe hypercholesterolemia.^{11,12}

Polygenic risk scores (PRSs), including those derived from common variants associated with CAD (PRS_{CAD})^{13,14} and Lp(a) levels [$PRS_{Lp(a)}$],^{15,16} may substantially complement monogenic testing by refining genetic risk and identifying a broader population at elevated genetic risk.^{17,18} The predictive performance of PRS_{CAD} and $PRS_{Lp(a)}$ has been consistently demonstrated in large cohorts, and their clinical utility recognized in scientific statements by the American Heart Association.^{17,19,20} However, they have not yet been incorporated into clinical guidelines.¹⁰ One major reason is the limited availability of data evaluating the integration and performance of genetic risk assessments, such as monogenic variants, PRS_{CAD} and $PRS_{Lp(a)}$ for CAD prediction, both independently and in conjunction with traditional clinical risk factors. Moreover, performance data across diverse ancestry groups remain extremely limited, yet are essential to ensure equity in risk stratification.²¹ Finally, a simple, clinically accessible genetic test that integrates these components is not yet readily available.

In the present study, we leveraged 2 large biobanks to develop, validate, and apply an integrated genetic risk model incorporating monogenic variants, PRS_{CAD} and $PRS_{Lp(a)}$, referred to as genetic probability for CAD ($GenProb_{CAD}$). We also evaluated its independent predictive performance for incident CAD beyond established clinical risk factors in both European and non-European populations.

METHODS

Data are available upon reasonable request from the corresponding authors.

Study Subjects

The study design for developing and validating $GenProb_{CAD}$ in 2 biobanks is summarized in the flowchart (Figure 1). The first is the UK Biobank (UKB), a population-based cohort comprising ≈500 000 individuals across the United Kingdom, aged 40 to 69 years at recruitment, with extensive phenotypic and genomic data available.²² Following the approach of Berry et al,¹⁹ CAD diagnoses were identified using *International Classification of Diseases, Tenth Revision* codes (Table S1), obtained from self-reported data and inpatient hospital records, with the last data access on June 12, 2023. Among participants receiving lipid-lowering therapy (LLT), adjusted low-density lipoprotein cholesterol (LDL-C) levels were calculated by dividing the measured LDL-C value by 0.7.^{11,19} Genetic data from whole-exome sequencing and SNP array (as well as imputed SNPs) were available. Family history of CAD was obtained through self-reported history of heart disease in either parent. The large sample size of the UKB enabled the development and validation of comprehensive genetic risk assessment models. We randomly selected 50% of participants to test the associations between genetic risk factors and CAD and to develop the $GenProb_{CAD}$ model. The remaining 50% of UKB participants were used for validation. This research has been conducted using the UKB Resource under Application Number 50295. All participants provided informed consent, and all data and samples were anonymized.

The second biobank was the Genomic Health Initiative (GHI), a US community-based health care biobank at Endeavor Health in Illinois.^{23,24} The same *International Classification of Diseases, Tenth Revision* codes (and *International Classification of Diseases, Ninth Revision* analogs) defined in the UKB were used to identify patients with CAD (Table S1). Germline DNA was isolated from blood samples and sequenced using low-coverage whole-genome sequencing blended with either high-coverage panel or whole-exome sequencing. SNPs were further imputed based on the 1000 Genomes Phase 3 haplotype reference panel, as previously described.²⁵ Family history of CAD was extracted from electronic health records using relevant *International Classification of Diseases* codes (Z82.49, V17.3) and keyword searches of categorical data and additional physician clinical documentation when available. The family history data obtained were then manually reviewed by A.J.M. and A.R.S. GHI was approved by the institutional review board of Endeavor Health. The UK Biobank has ethical approval from the North West Multicenter Research Ethics Committee. The

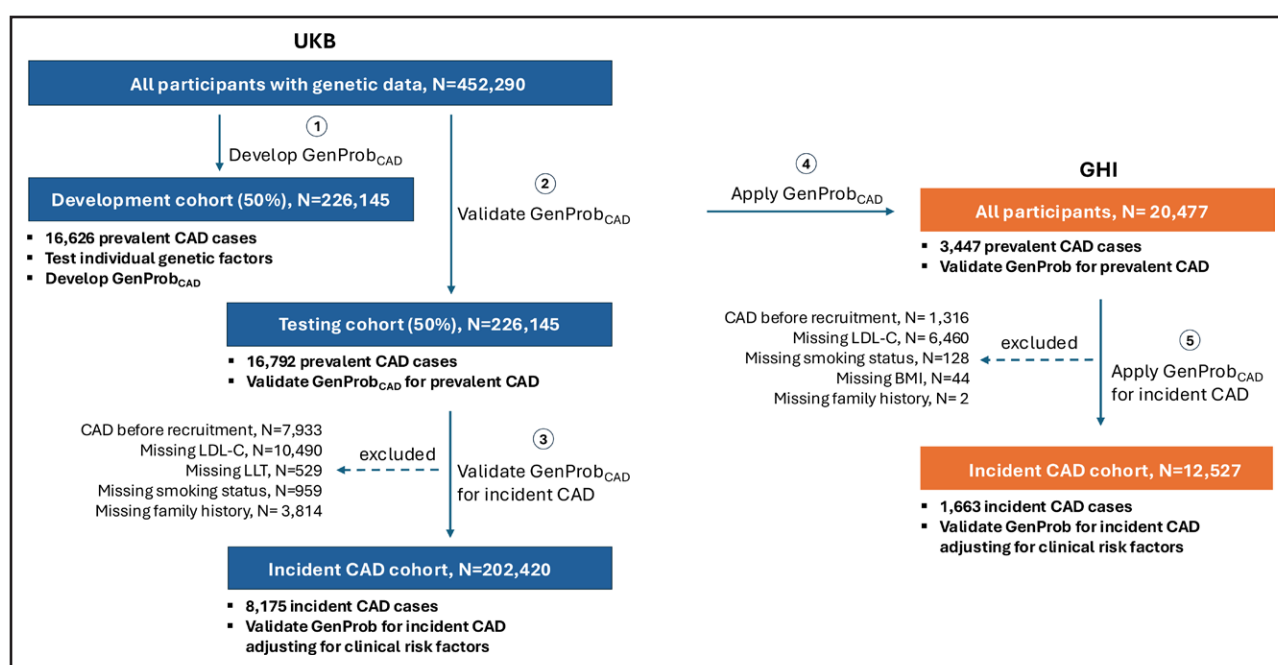


Figure 1. Flowchart of study design.

Briefly, we (1) tested the association of major genetic risk factors with all coronary artery disease (CAD) in a randomly selected 50% subset of UK Biobank (UKB) participants with available genetic data, and developed a genetic probability (GenProb_{CAD}) model that included all significant genetic risk factors using Cox regression; (2) validated the performance of GenProb_{CAD} for predicting all CAD in the remaining 50% of UKB participants; (3) assessed the independent performance of GenProb_{CAD} for predicting incident CAD after recruitment in the UKB validation cohort, adjusting for major clinical risk factors at the time of recruitment, including low-density lipoprotein cholesterol (LDL-C) levels, family history for CAD, smoking status, diabetes, obesity, hypertension, and lipid-lowering therapy (LLT); (4) applied GenProb_{CAD} for predicting all CAD in the Genomic Health Initiative (GHI), a community-based health care biobank with diverse ancestry populations; and (5) evaluated the performance of GenProb_{CAD} for predicting incident CAD after recruitment in GHI, also adjusting for major clinical risk factors at recruitment.

GHI has been approved under Endeavor Health's institutional review board. All the participants provided informed consent to participate in GHI.

Genetic Ancestry

Genetic ancestry probabilities for each individual in the UKB and GHI cohorts were estimated using principal components derived from 16309 ancestry-informative markers distributed across the genome. Ancestral populations included Finnish, Non-Finnish European, Ashkenazi Jewish, East Asian, African, Admixed American, and South Asian. Individuals were assigned to a specific ancestry group if their predicted probability for that ancestry exceeded 75%. For downstream analyses, participants were classified as having European ancestry if >75% of their genetic ancestry was Finnish, Non-Finnish European, and Ashkenazi Jewish, and as having non-European ancestry otherwise.

Annotation of PVs in Monogenic Genes

All candidate protein altered variants in exons, exon-intron junctions, and untranslated regions of *LDLR*, *APOB*, and *PCSK9* were identified from sequencing data. Their pathogenicity was annotated based on the criteria recommended by the ClinGen Familial Hypercholesterolemia Expert Panel Specifications to the ACMG/AMP Variant Classification Guidelines, Version 1.2 by Z.S.²⁶ We identified 176 pathogenic and likely PVs in these genes, which are grouped as PVs. These PVs, together with specific variant classification criteria, are listed in Table S2.

PRS Selection and Calculation

Three PRSs were evaluated in the study, including PRS_{CAD}, PRS_{Lp(a)}, and PRS_{HCL} for hypercholesterolemia. Each PRS was selected from multiple available PRS methods in the PGS Catalog based on larger sample sizes and broader ancestry representation for (1) the source GWAS (genome-wide association study), (2) score development and training, and (3) PGS validation, as well as minimal contribution from the UKB to reduce potential overfitting.²⁷ For PRS_{CAD}, PGS003725 (1 296 172 SNPs) was selected. This score was derived from GWAS summary statistics of 1 443 386 individuals of diverse ancestries from 11 large cohorts, developed using the European ancestry subset of the UKB, and validated in multiethnic populations.¹³ Furthermore, this PRS was derived from genome-wide summary statistics that include SNPs associated not only with CAD itself but also with related cardiometabolic traits such as BMI, peripheral artery disease, stroke, kidney function (GFR), diabetes, blood pressure, and lipid levels (LDL-C, HDL-C, and triglycerides). For PRS_{Lp(a)}, PGS000667 (43 SNPs) was selected. It was based on GWAS summary statistics of 48 333 individuals from 5 large cohorts of various ancestries, with model development and validation performed in 15 multiethnic study populations.¹⁵ For PRS_{HCL}, PGS002764 (1 091 280 SNPs) was selected. This model was developed and validated among 340 951 individuals from the FinnGen study in Finland.²⁸ For each PRS, raw scores were calculated for each individual using the provided scoring files and their genotype data. Ancestry-adjusted PRS values were then derived by adjusting for the first

4 principal components, as previously described.²⁹ To categorize the continuous PRS, the relative risk (RR) associated with each percentile of PRS_{CAD} and PRS_{HCL} was estimated using the entire UKB cohort as the reference population. Two percentile cutoffs corresponding to estimated RRs of 0.5 and 1.5 were identified. Based on these thresholds, individuals were categorized into 3 genetic risk groups: low- (RR <0.5), average- (RR 0.5–1.49), and high-risk (≥ 1.5).

Statistical Methods

Associations between genetic risk factors and CAD risk were tested using multivariable Cox proportional hazards models adjusting for relevant covariates. Hazard ratios (HRs) and 95% CIs were estimated. Interactions among genetic risk factors were also tested. For the analysis including all CAD cases, the timescale was defined from birth to the age at first diagnosis for cases, and to the age at the most recent follow-up for controls. Covariates included demographic variables (age, sex, and genetic background). For incident CAD after recruitment, the timescale was defined from the date of recruitment to the date of CAD diagnosis for cases, or to the most recent follow-up for controls. Covariates for this analysis included demographic variables and baseline clinical risk factors at recruitment (LDL-C levels, family history of CAD, smoking status, diabetes, obesity, hypertension, and LLT).

GenProb_{CAD} for CAD was constructed by incorporating all significant genetic risk factors into a Cox proportional hazards regression. It was calculated based on the linear predictor from the Cox model. Following the approach described by Fahed et al,¹⁸ the probability of developing CAD by age 75 years (used here as a pragmatic age-based summary of risk) was estimated using the survivor function from the Cox model based on significant genetic risk factors. Similar to PRS, the continuous GenProb_{CAD} was also categorized into 3 risk groups based on 2 percentile cutoffs corresponding to estimated RRs of 0.5 and 1.5 (<0.5, 0.5–1.49, and ≥ 1.5).

The predictive performance of each genetic risk factor and GenProb_{CAD} for CAD was assessed using the concordance index (C-index), as well as calibration plots comparing deciles of predicted CAD risk by age 75 years, estimated using the UKB baseline hazard, with the corresponding observed CAD prevalence. In addition, for visual presentation, age-based time-to-event analyses were used to estimate observed CAD prevalence across genetic risk strata, with age as the underlying time scale and death treated as a censoring event. Group differences were assessed using log-rank tests.

RESULTS

Study Subjects

All participants with available genetic data were included in the analysis, including 452 290 individuals from the UKB and 20 477 from the GHI (Figure 1 and Table S3). The median age at last follow-up was 72.04 years (interquartile range, 64.62–77.37) in the UKB and 63.95 years (interquartile range, 50.32–74.25) in the GHI. Higher proportions of female participants were in both UKB (54.24%) and GHI (64.61%). The

majority of UKB participants (93.93%) were of European ancestry, whereas 26.99% of GHI participants were of other ancestries. The CAD prevalence was lower in the population-based UKB (7.39%) than in the community-based GHI (16.83%). The incident of CAD after recruitment was 4.04% in the UKB and 13.28% in the GHI after a median follow-up of 14.24 and 7.72 years, respectively.

Association of Genetic Risk Factors With CAD Risk in the UKB Development Cohort

In the UKB development cohort, PVs, PRS_{CAD} , and $PRS_{LP(a)}$ were each independently and significantly associated with increased risk of CAD in multivariable analysis adjusting for age, gender, and genetic background, all $P < 0.001$ (Table 1). Specifically, among 226 146 subjects, 740 (0.33%) carried a PV in 1 of 3 genes (*LDLR*, *APOB*, and *PCSK9*), with an HR of 2.55 (95% CI, 2.12–3.06) compared with noncarriers. PRS_{CAD} classified many more high-risk subjects ($N = 27\,071$, 11.97%), with an HR of 2.82 (95% CI, 2.73–2.92) compared with those at average risk. $PRS_{LP(a)}$ also classified a large proportion of individuals as high risk ($N = 47\,687$; 21.09%), but their CAD risk was more modest (HR, 1.27 [95% CI, 1.19–1.34]). Among the 47 687 individuals classified as high risk by $PRS_{LP(a)}$, the vast majority ($N = 39\,784$) were classified as low or average risk by PRS_{CAD} alone, indicating that $PRS_{LP(a)}$ identified a large group of additional individuals at increased genetic risk who would not have been captured by PRS_{CAD} -based approaches. In contrast, PRS_{HCL} was not significantly associated with CAD. No significant interactions were observed among the 3 significant risk factors, $P > 0.05$.

An integrated model, GenProb_{CAD}, was constructed using a Cox regression model that incorporated the 3 significant genetic risk factors: $GenProb_{CAD} = 0.878 \times PV + 0.582 \times PRS_{CAD} + 0.096 \times PRS_{LP(a)}$. The model demonstrated strong calibration, with predicted risk closely matching observed CAD risk (Figure S1A). GenProb_{CAD}-derived risk estimates by age 75 years demonstrated a significantly higher C statistic (95% CI) of 0.73 (0.73–0.74) compared with PVs of 0.68 (0.68–0.69), PRS_{CAD} of 0.72 (0.72–0.73), and $PRS_{LP(a)}$ of 0.69 (0.69–0.7), all $P < 0.05$ (Table 1; Figure S2A). Notably, GenProb_{CAD} classified substantially more individuals as high risk (16%) than monogenic PVs (0.33%), while identifying individuals with a similar CAD proportion (15.34% versus 15.81%, respectively). The mean age at CAD diagnosis was 7.47 years younger in individuals with high versus low GenProb_{CAD}, 48.59 and 56.06 years, respectively, $P < 0.05$. In contrast, the difference between monogenic PV carriers and noncarriers was only 2.98 years, 48.34 and 51.32 years, respectively, $P = 0.12$.

Table 1. Association of Genetic Risk Factors With CAD Risk in the UK Biobank

Risk factors	Subjects, N (%)	CAD, N (%)	Age dx, mean (95% CI)	HR (95% CI)*	P value*	C-index (95% CI)	P value
Development cohort, N=226 145							
Carriers						0.68 (0.68–0.69)	<1×10 ⁻³⁰⁰
No	225 405 (99.67)	16 509 (7.32)	51.32 (50.99–51.65)	1.00 (reference)			
Yes	740 (0.33)	117 (15.81)	48.34 (44.55–52.14)	2.55 (2.12–3.06)	7.04×10 ⁻²⁴		
PGS _{CAD} †						0.72 (0.72–0.73)	<1×10 ⁻³⁰⁰
<0.5	13 622 (6.02)	331 (2.43)	56.83 (54.53–59.13)	0.36 (0.32–0.40)	3.30×10 ⁻⁷⁶		
0.5–1.5	185 452 (82.01)	11 936 (6.44)	52.22 (51.83–52.61)	1.00 (reference)			
≥1.5	27 071 (11.97)	4359 (16.1)	48.36 (47.72–48.99)	2.82 (2.73–2.92)	<1×10 ⁻³⁰⁰		
PGS _{Lp(a)} ‡						0.69 (0.69–0.7)	<1×10 ⁻³⁰⁰
<0.5	158 918 (70.27)	10 577 (6.66)	51.58 (51.17–51.99)	0.85 (0.81–0.90)	1.65×10 ⁻⁰⁸		
0.5–1.5	19 540 (8.64)	1486 (7.6)	50.71 (49.6–51.81)	1.00 (reference)			
≥1.5	47 687 (21.09)	4563 (9.57)	50.83 (50.21–51.46)	1.27 (1.19–1.34)	6.06×10 ⁻¹⁵		
PGS _{HCL} §							
<0.5	20 668 (9.14)	1083 (5.24)	51.75 (50.43–53.07)	0.70 (0.66–0.75)	7.32×10 ⁻²⁹		
0.5–1.5	178 725 (79.03)	12 947 (7.24)	51.49 (51.12–51.86)	1.00 (reference)			
≥1.5	26 752 (11.83)	2596 (9.7)	50.17 (49.34–51.01)	1.38 (1.33–1.44)	2.99×10 ⁻⁵¹		
GenProb _{CAD} ¶						0.73 (0.73–0.74)	Reference
<0.5	29 398 (13)	807 (2.75)	56.06 (54.59–57.53)	0.41 (0.38–0.44)	1.04×10 ⁻¹³³		
0.5–1.5	160 563 (71)	10 270 (6.4)	52.39 (51.97–52.81)	1.00 (reference)			
≥1.5	36 184 (16)	5549 (15.34)	48.59 (48.02–49.15)	2.67 (2.59–2.76)	<1×10 ⁻³⁰⁰		
Validation cohort, N=226 145							
Carriers						0.68 (0.68–0.69)	<1×10 ⁻³⁰⁰
No	225 361 (99.65)	16 671 (7.4)	51.36 (51.03–51.69)	1.00 (reference)			
Yes	784 (0.35)	121 (15.43)	47.61 (44.1–51.12)	2.32 (1.94–2.77)	3.64×10 ⁻²⁰		
PGS _{CAD} †						0.72 (0.72–0.73)	<1×10 ⁻³⁰⁰
<0.5	13 545 (5.99)	292 (2.16)	55.9 (53.51–58.28)	0.32 (0.28–0.36)	5.03×10 ⁻⁸⁴		
0.5–1.5	185 384 (81.98)	11 975 (6.46)	52.35 (51.96–52.73)	1.00 (reference)			
≥1.5	27 216 (12.03)	4525 (16.63)	48.36 (47.73–48.98)	2.91 (2.81–3.01)	<1×10 ⁻³⁰⁰		
PGS _{Lp(a)} ‡						0.69 (0.69–0.7)	<1×10 ⁻³⁰⁰
<0.5	159 016 (70.32)	10 783 (6.78)	51.53 (51.12–51.93)	0.88 (0.83–0.93)	5.68×10 ⁻⁰⁶		
0.5–1.5	19 965 (8.83)	1506 (7.54)	51.84 (50.77–52.92)	1.00 (reference)			
≥1.5	47 164 (20.86)	4503 (9.55)	50.7 (50.07–51.33)	1.28 (1.21–1.36)	2.68×10 ⁻¹⁶		
GenProb _{CAD} ¶						0.73 (0.73–0.74)	Reference
<0.5	29 096 (12.87)	753 (2.59)	55.96 (54.45–57.47)	0.38 (0.36–0.41)	1.13×10 ⁻¹⁴¹		
0.5–1.5	160 858 (71.13)	10 395 (6.46)	52.39 (51.98–52.81)	1.00 (reference)			
≥1.5	36 191 (16)	5644 (15.6)	48.76 (48.2–49.32)	2.68 (2.59–2.77)	<1×10 ⁻³⁰⁰		

CAD indicates, coronary artery disease; HR, hazard ratio; PGS, polygenic risk score; RR, relative risk; and UKB, UK Biobank.

*Adjusted for age, sex, and the first 10 principal components of ancestry.

†Cutoffs are -1.558855 and 1.175333 for 0.5 and 1.5 RR with the total UKB data.

‡Cutoffs are -0.000497 and 0.709967 for 0.5 and 1.5 RR with the total UKB data.

§Cutoffs are -1.34242 and 1.16498551988921 for 0.5 and 1.5 RR with the total UKB data.

¶GenProb at age 75 years with cutoffs of 0.0382527451727196 and 0.130423388954541 for 0.5 and 1.5 RR with the UKB development cohort.

Confirmation of GenProbCAD for Predicting CAD Risk in the Validation Cohort

The predictive performance of GenProb_{CAD} was confirmed in the remaining 50% of UKB participants, with predicted risks closely matching observed CAD prevalence (Figure S1B). This validation cohort demonstrated

results consistent with the discovery cohort, including HR, C statistic, the proportion of high-risk individuals, their corresponding CAD prevalence, and age at CAD diagnosis (Table 1; Figure S2B).

Kaplan-Meier type curves from age-based time-to-event analyses revealed several additional key features. First, while carriers of monogenic PVs had

a significantly higher proportion of observed CAD prevalence compared with noncarriers, $P<0.001$ (Figure 2B), their CAD prevalence was substantially modified by PRS_{CAD} (Figure 2B). Notably, a crossover was observed in which noncarriers with high PRS_{CAD} had a higher CAD prevalence than carriers with average or low PRS_{CAD} . This pattern remained significant after adjusting for demographic covariates in Cox regression models. Similarly, $PRS_{Lp(a)}$ modified CAD prevalence in both carriers and noncarriers (Figure 2C), although no crossover pattern was observed. Second, $GenProb_{CAD}$ provided the most effective stratification of CAD prevalence, identifying substantial proportions of individuals at both high (16.00%) and low (12.87%) genetic risk, with marked differences in time to CAD diagnosis (Figure 2D).

Finally, a consistent and clinically meaningful observation was the earlier onset of CAD among individuals identified as high risk by genetic factors. $GenProb_{CAD}$ was particularly informative in stratifying risk for premature CAD, defined as diagnosis before age 55 in men and age 60 in women (Figure 2E). The proportion of premature CAD was significantly higher in the high $GenProb_{CAD}$ group (49.8%) compared with the low $GenProb_{CAD}$ group (33.7%), representing a 16.1% absolute difference ($P<0.001$). The corresponding proportions were 54.6% among monogenic PV carriers and 43.7% among noncarriers, a 10.9% difference ($P<0.001$).

Performance of GenProbCAD for Predicting Incident CAD Adjusting for Clinical Risk Factors

To assess the independent performance of $GenProb_{CAD}$ in predicting CAD risk from clinical risk factors, we constructed a subcohort within the UKB validation set by excluding individuals diagnosed with CAD before recruitment and collected information for other clinical risk factors at recruitment, including LDL-C levels, family history of CAD, smoking status, diabetes, obesity, hypertension, and LLT (Figure 1).

In the multivariable model including clinical risk factors alone within this subcohort, several established risk factors were significantly associated with incident CAD, including male sex, older age, family history of CAD, smoking, hypertension, diabetes, obesity, and elevated LDL-C (Table S4).

When $GenProb_{CAD}$ was added to the model, it remained a significant predictor of incident CAD after adjusting for demographic and clinical risk factors. Compared with the average-risk group, individuals classified as high-risk by $GenProb_{CAD}$ had an HR (95% CI) of 2.10 (2.00–2.21), $P<0.001$ (Figure 3A). Furthermore, $GenProb_{CAD}$ identified additional individuals at lower risk for incident CAD, with an HR of 0.51 (0.46–0.56), $P<0.001$. Notably, the HR of $GenProb_{CAD}$ was higher

than the HR for monogenic PV carriers, which was not statistically significant at 1.22 (0.90–1.66), $P=0.20$ (Figure 3B).

Individuals with high-risk $GenProb_{CAD}$ exhibited consistently elevated CAD incidence across multiple adjusted LDL-C strata. Strikingly, even among those with LDL-C levels of 130 to 160 mg/dL, individuals with high $GenProb_{CAD}$ had a CAD incidence of 8.01%, exceeding that observed in individuals with LDL-C ≥ 250 mg/dL (7.32%) in the overall incident cohort (Figure 3C).

Application of GenProbCAD for Predicting CAD Risk in the GHI

To evaluate the generalizability of $GenProb_{CAD}$ in a real-world setting with broader ancestral diversity, we calculated $GenProb_{CAD}$ for participants in the GHI using the coefficients and cutoffs derived from the UKB. In the overall GHI population, the predicted risks from $GenProb_{CAD}$ closely matched observed CAD prevalence (Figure S1C). $GenProb_{CAD}$ was significantly associated with time to CAD diagnosis, $P<0.001$ (Table 2). Compared with monogenic PV carriers, $GenProb_{CAD}$ had a higher C statistic (0.78 versus 0.76), $P=0.01$ (Figure S2C) and classified a substantially greater proportion of individuals (15.67% versus 0.46%). The mean age at CAD diagnosis was also significantly lower among individuals in the high $GenProb_{CAD}$ risk group, $P<0.001$. A notable difference from the UKB was a lower HR estimate for $GenProb_{CAD}$ (1.83) compared with that of monogenic PV carriers (2.45).

In the GHI incident subcohort constructed by excluding individuals diagnosed with CAD before recruitment (Figure 1), the independent predictive performance of $GenProb_{CAD}$ beyond clinical risk factors was also demonstrated (Table 2). Compared with the average $GenProb_{CAD}$, individuals classified as high $GenProb_{CAD}$ had an HR of 1.49 (95% CI, 1.32–1.69), and those with low $GenProb_{CAD}$ had an HR of 0.70 (95% CI, 0.59–0.82), both $P<0.001$. In comparison, the HR was 1.76 (0.99–3.11) for monogenic PV carriers, $P=0.05$. Similar results were found in both males and female in the sex-stratified analyses (Table S5).

Evaluation of key factors related to genetic testing strategies revealed several findings with practical implications (Table S6). First, only a small proportion of participants (7.49%) in the GHI incident subcohort had LDL-C levels ≥ 190 mg/dL (Figure 4A). Second, among these participants, only 20 PV carriers were identified (Figure 4B). More PV carriers ($N=41$) were observed among participants with lower LDL-C levels, though at reduced carrier rates. In comparison, $GenProb_{CAD}$ classified 182 participants with LDL-C ≥ 190 mg/dL as high risk and an even larger number ($N=1784$) among those with lower LDL-C levels. Third, incident CAD proportions were similar across PV carriers and high-risk $GenProb_{CAD}$

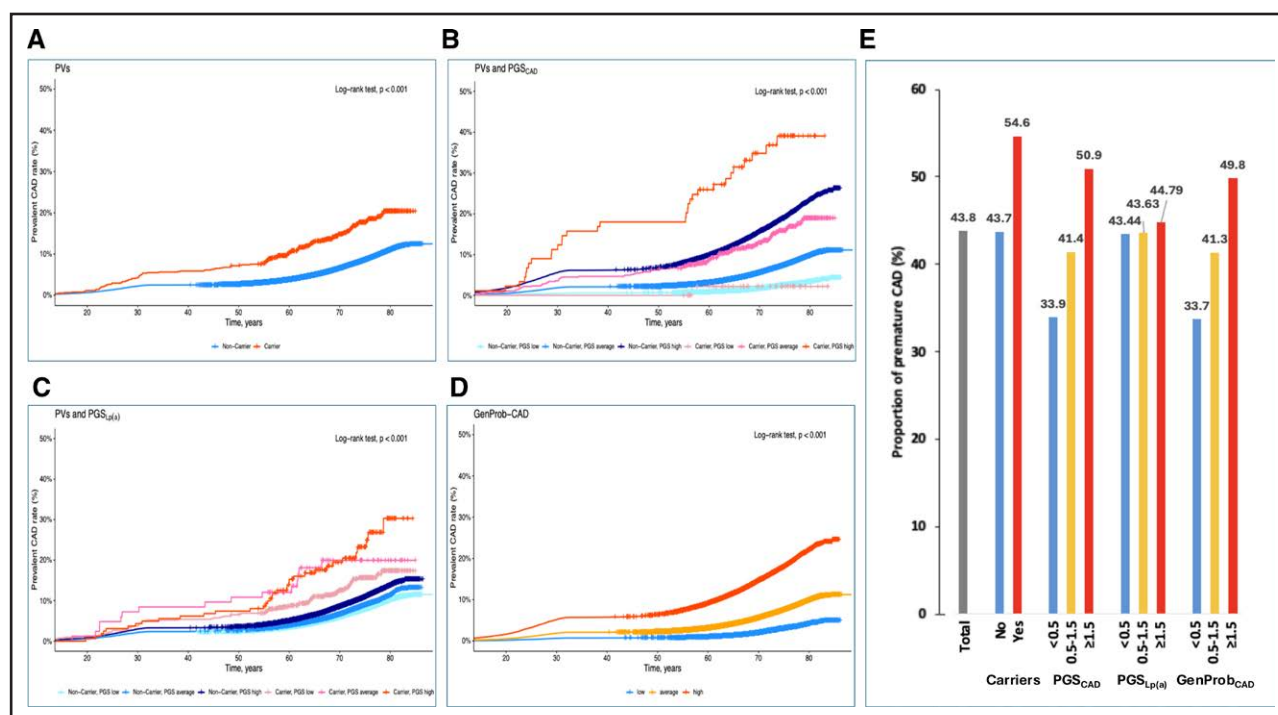


Figure 2. Association of genetic risk factors with observed coronary artery disease (CAD) prevalence in the UK Biobank validation cohort.

Observed CAD prevalence, estimated using Kaplan-Meier methods from age-based time-to-event analyses, is shown by: (A) pathogenic variant (PV) carrier status; (B) PV carrier status and polygenic risk score with CAD (PRS_{CAD}); (C) PV carrier status and polygenic risk score with Lp(a); lipoprotein(a) [$PRS_{Lp(a)}$]; and (D) integrated genetic risk categories defined by $GenProb_{CAD}$. The proportion of premature CAD, defined as diagnosis before age 55 years in men and before age 60 years in women, by genetic risk factors is shown in (E).

groups (16.5% to 25%) regardless of LDL-C strata (Figure 4C). These proportions were comparable to individuals with LDL-C ≥190 mg/dL (18.34%), but significantly higher than those with LDL-C <130 mg/dL (12.29%) or 130–189 mg/dL (13.95%), $P < 0.001$. Overall, a universal genetic testing strategy using $GenProb_{CAD}$ identified 1966 high-risk participants in the incident cohort, of whom 345 developed CAD, accounting for 20.75% of all incident CAD cases ($N=1663$). This yield was 69-fold higher than that of the current monogenic-only testing strategy focused on patients with severe hypercholesterolemia, which identified only 20 PV carriers (5 developed CAD), accounting for 0.3% of all incident CAD.

Importantly, the predictive performance of $GenProb_{CAD}$ for CAD risk was generally consistent in both European and non-European ancestry populations (Table 3). In each ancestry population, $GenProb_{CAD}$ was significantly associated with time to CAD diagnosis, $P < 0.001$. Compared with monogenic PVs, $GenProb_{CAD}$ demonstrated higher C statistics (Figure S2D and S2E) and identified a substantially larger proportion of individuals at high genetic risk. These high-risk individuals exhibited an earlier age at diagnosis and had a CAD risk comparable to that of monogenic PV carriers. In the incident subcohorts, $GenProb_{CAD}$ remained an independent predictor of incident CAD after adjusting for clinical risk factors in individuals of European ancestry.

A similar trend was observed in the non-European population, although the association did not reach statistical significance.

DISCUSSION

Despite substantial differences between the 2 biobanks in recruitment setting (population-based versus health care-based), geographic region (United Kingdom versus United States), ancestry composition (predominantly European versus ancestrally diverse), and CAD prevalence and incidence (much higher in GHI, likely reflecting its health care-based recruitment, more complete data capture from electronic health records, and potential coding differences between the United States and United Kingdom), several consistent findings emerged from this study. First, 3 known genetic risk factors, monogenic PVs, PRS_{CAD} and $PRS_{Lp(a)}$, were each significantly and independently associated with risk to CAD. Second, the risk associated with guideline-recommended monogenic PVs was substantially reclassified by PRS_{CAD} , as demonstrated by a crossover in CAD proportion between noncarriers with high-risk PRS_{CAD} group and carriers with low-risk PRS_{CAD} group. Third, integration of the 3 factors into a single model ($GenProb_{CAD}$) outperformed each individual component in assessing genetic risk for CAD. Specifically, $GenProb_{CAD}$ identified 46 times more

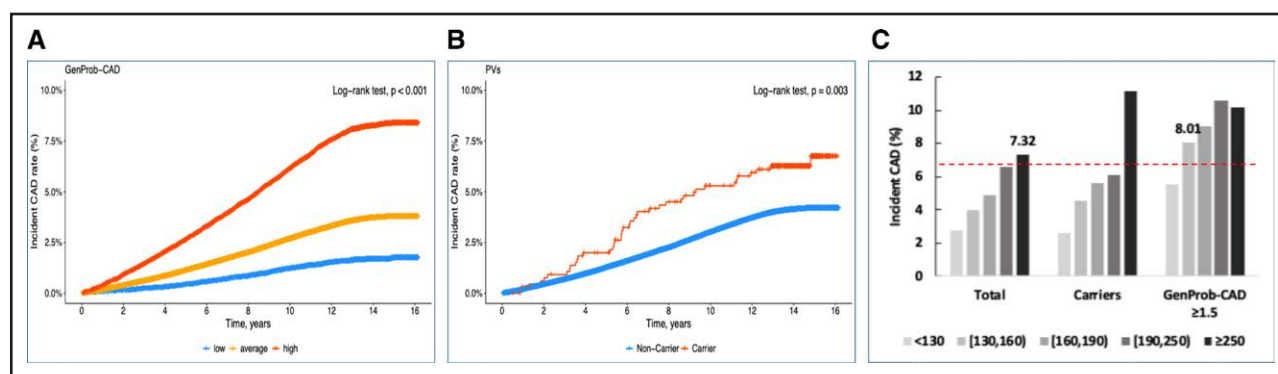


Figure 3. Association of genetic risk factors with incident coronary artery disease (CAD) after adjusting for demographic and clinical risk factors (low-density lipoprotein cholesterol [LDL-C] levels, family history of CAD, smoking status, diabetes, obesity, hypertension, and lipid-lowering therapy [LLT]) at recruitment in the subcohort of the UK Biobank (UKB) validation set.

Kaplan-Meier curves for time to incident CAD diagnosis are shown by (A) $\text{GenProb}_{\text{CAD}}$ and (B) pathogenic variant (PV) status. CAD incidence across multiple adjusted LDL-C strata in the entire subcohort, carriers of PVs, and high-risk $\text{GenProb}_{\text{CAD}}$ is shown in (C) where the red horizontal dotted line indicates the proportion of CAD among individuals with LDL-C ≥ 250 mg/dL (7.32%) in the overall incident cohort.

individuals in the population at high genetic risk, with CAD proportions exceeding or comparable to that of monogenic PV carriers. Fourth, 49.8% of the CAD patients in the high $\text{GenProb}_{\text{CAD}}$ group had premature CAD, significantly higher than that of low $\text{GenProb}_{\text{CAD}}$ group (33.7%). Fifth, the predictive performance of $\text{GenProb}_{\text{CAD}}$ was independent of established demographic and clinical risk factors. Notably, individuals with high $\text{GenProb}_{\text{CAD}}$ had consistently higher CAD incidence across multiple LDL-C levels, including those with adjusted LDL-C levels of 130 to 160 mg/dL. Finally, the enhanced performance of $\text{GenProb}_{\text{CAD}}$ was consistent across both European and non-European ancestry groups.

These findings, along with evidence from other published studies, underscore the need to revise the current genetic testing strategy for FH and CAD, which has historically relied on monogenic testing among patients suspected of having FH.¹⁰ This approach originated from early studies in clinically diagnosed FH populations and highly selected case-control cohorts, where higher carrier frequencies and stronger associations with CAD were observed. However, contemporary data from large-scale sequencing studies in unselected populations consistently show that monogenic FH PVs are rare in the general population (0.3–0.4%), present in both individuals with and without FH, and only modestly enriched among patients with severe hypercholesterolemia ($<2\%$).^{6,11,30,31} The risk conferred by these PVs is lower than previously estimated; more than half of monogenic PV carriers have LDL-C levels below 190 mg/dL, and their risk for CAD is approximately 2-fold compared with noncarriers.³²

Therefore, supplementing monogenic testing with PRS, as implemented in integrated models such as $\text{GenProb}_{\text{CAD}}$, addresses the limitations of monogenic-only testing and offers a more comprehensive and accurate approach to genetic risk assessment. Moreover, eligibility for genetic testing should be expanded

beyond individuals with severe hypercholesterolemia to include the general population, as high $\text{GenProb}_{\text{CAD}}$ is associated with elevated CAD risk across all LDL-C levels, including among individuals with LDL-C <190 mg/dL. This has important public health implications, given that the vast majority of individuals in the general population fall into this LDL-C category. Using the GHI incident subcohort as an example, only 7.49% had severe hypercholesterolemia (LDL-C ≥ 190 mg/dL). If testing were restricted to this group and limited to monogenic variants, only 20 PV carriers would have been identified, of whom 5 developed CAD, representing just 0.3% of all incident CAD cases (N=1663). In comparison, $\text{GenProb}_{\text{CAD}}$ identified 1966 high-risk participants, of whom 345 developed CAD, accounting for 20.75% of all incident cases. Overall, a universal genetic testing strategy using $\text{GenProb}_{\text{CAD}}$ achieved a 69-fold greater yield than current PV-only testing of severe hypercholesterolemia.

The need for integrated monogenic and polygenic risk assessment is further emphasized by the substantial reclassification of risk based on PRSs, as evidenced by the crossover in CAD proportion between carriers in the low-PRS group and noncarriers in the high-PRS group. This integrated approach allows for more accurate risk stratification of monogenic PV carriers and enables tailored and timely intervention strategies. More importantly, it reveals high-risk individuals among noncarriers ($\approx 12\%$ in the GHI cohort) who might otherwise receive false reassurance with current clinical risk assessments. Recognizing that their CAD risk may exceed that of some monogenic PV carriers could motivate these individuals to adopt more aggressive preventive measures, such as LLT, lifestyle modification, and early atherosclerosis detection using coronary artery calcium scoring.^{2,33} Finally, $\text{GenProb}_{\text{CAD}}$ offers practical advantages by providing a single, integrated measure of genetic risk, simplifying interpretation and communication compared with

Table 2. Association of Genetic Risk Factors With CAD Risk in the GHI

Risk factors	Subjects, N (%)	CAD, N (%)	Age dx, mean (95% CI)	HR (95% CI)*	P value*	C-index (95% CI)	P value
Entire cohort, N=20 477							
Carriers						0.76 (0.76–0.77)	0.01
No	20382 (99.54)	3418 (16.77)	68 (67.63–68.36)	1.00 (reference)			
Yes	95 (0.46)	29 (30.53)	62.97 (59–66.93)	2.45 (1.70–3.54)	1.60×10 ^{−06}		
PGS _{CAD} †						0.78 (0.77–0.78)	0.24
<0.5	1417 (6.92)	158 (11.15)	71.84 (70.28–73.41)	0.50 (0.43–0.59)	1.00×10 ^{−16}		
0.5–1.5	16655 (81.34)	2712 (16.28)	68.36 (67.95–68.77)	1.00 (reference)			
≥1.5	2405 (11.74)	577 (23.99)	64.96 (64.05–65.87)	1.93 (1.76–2.11)	1.05×10 ^{−45}		
PGS _{Lp(a)} ‡						0.76 (0.76–0.77)	0.01
<0.5	14408 (70.36)	2355 (16.35)	68.48 (68.04–68.91)	0.87 (0.78–0.96)	0.00561654		
0.5–1.5	2764 (13.5)	482 (17.44)	66.28 (65.26–67.29)	1.00 (reference)			
≥1.5	3305 (16.14)	610 (18.46)	67.25 (66.34–68.16)	1.09 (0.97–1.24)	0.14945374		
GenProb _{CAD} §						0.78 (0.77–0.79)	Reference
<0.5	2886 (14.09)	337 (11.68)	71.66 (70.59–72.73)	0.55 (0.49–0.62)	3.20×10 ^{−24}		
0.5–1.5	14382 (70.23)	2349 (16.33)	68.25 (67.81–68.68)	1.00 (reference)			
≥1.5	3209 (15.67)	761 (23.71)	65.41 (64.6–66.22)	1.83 (1.68–1.98)	2.56×10 ^{−46}		
Incident subcohort, N=12527							
Carriers						0.77 (0.76–0.79)	1.93×10 ^{−03}
No	12466 (99.51)	1651 (13.24)	69.53 (69.02–70.03)	1.00 (reference)			
Yes	61 (0.49)	12 (19.67)	63.5 (58.71–68.29)	1.76 (0.99–3.11)	0.05		
GenProb _{CAD} §						0.78 (0.77–0.79)	Reference
<0.5	1676 (13.38)	171 (10.2)	73.13 (71.71–74.55)	0.70 (0.59–0.82)	1.43×10 ^{−05}		
0.5–1.5	8885 (70.93)	1147 (12.91)	69.73 (69.12–70.34)	1.00 (reference)			
≥1.5	1966 (15.69)	345 (17.55)	66.85 (65.73–67.97)	1.49 (1.32–1.69)	2.29×10 ^{−10}		

CAD indicates, coronary artery disease; GHI, Genomic Health Initiative; RR, ; HR, hazard ratio; PGS, polygenic risk score; relative risk; and UKB, UK Biobank.
*Adjusted for age, sex, and the first 10 principal components of ancestry.
†Cutoffs are −1.558855 and 1.175333 for 0.5 and 1.5 RR with the total UKB data.
‡Cutoffs are −0.000497 and 0.709967 for 0.5 and 1.5 RR with the total UKB data.
§GenProb at age 75 years with cutoffs of 0.0382527451727196 and 0.130423388954541 for 0.5 and 1.5 RR with the UKB development cohort.

reporting separate results for monogenic variants and PRSs, which may differ and be challenging to reconcile for both physicians and patients. In terms of timing for genetic testing, the substantially higher proportion of individuals with high GenProb_{CAD} diagnosed with CAD at a younger age highlights the

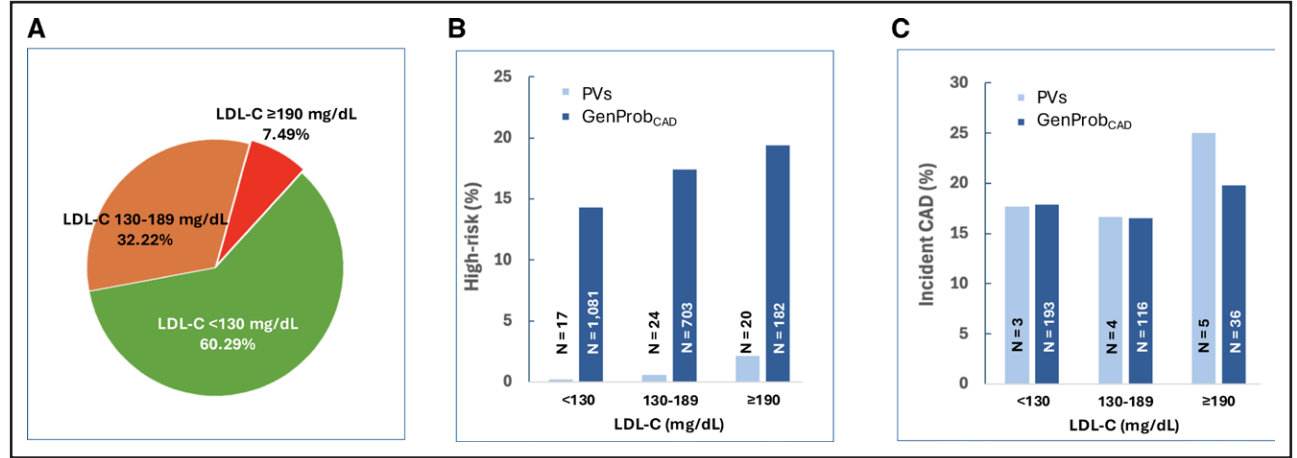


Figure 4. Comparison of genetic testing strategies in the Genomic Health Initiative (GHI) incident cohort.
A, Distribution of adjusted low-density lipoprotein cholesterol (LDL-C) levels at recruitment. **B**, Proportion of high-risk individuals classified by pathogenic variants (PVs) and by GenProb_{CAD} across LDL-C strata. **C**, Proportion of incident coronary artery disease (CAD) among high-risk individuals defined by PVs and by GenProb_{CAD} across LDL-C strata.

Table 3. Association of Genetic Risk Factors With CAD Risk in the GHI

Risk factors	Subjects, N (%)	CAD, N (%)	Age dx, mean (95% CI)	HR (95% CI)*	P value*	C-index (95% CI)	P value
European ancestry, N=14 951							
Carriers						0.76 (0.75–0.77)	0.01
No	14 890 (99.59)	2751 (18.48)	68.77 (68.37–69.16)	1.00 (reference)			
Yes	61 (0.41)	23 (37.7)	62.43 (57.67–67.2)	2.63 (1.74–3.96)	4.21×10 ^{−06}		
PGS _{CAD} [†]						0.77 (0.77–0.78)	0.18
<0.5	1215 (8.13)	146 (12.02)	72.54 (70.96–74.12)	0.51 (0.43–0.60)	2.02×10 ^{−15}		
0.5–1.5	12 069 (80.72)	2179 (18.05)	69.1 (68.67–69.54)	1.00 (reference)			
≥1.5	1667 (11.15)	449 (26.93)	65.58 (64.57–66.59)	1.98 (1.79–2.20)	8.28×10 ^{−39}		
PGS _{Lp(a)} [‡]						0.76 (0.75–0.77)	0.01
<0.5	10 940 (73.17)	1930 (17.64)	69.03 (68.56–69.5)	0.85 (0.76–0.96)	0.01		
0.5–1.5	1636 (10.94)	344 (21.03)	67.66 (66.51–68.81)	1.00 (reference)			
≥1.5	2375 (15.89)	500 (21.05)	68.22 (67.28–69.17)	1.12 (0.97–1.28)	0.12		
GenProb _{CAD} [§]						0.78 (0.77–0.79)	Reference
<0.5	2379 (15.91)	305 (12.82)	71.96 (70.84–73.08)	0.55 (0.49–0.62)	1.03×10 ^{−21}		
0.5–1.5	10 346 (69.2)	1874 (18.11)	68.99 (68.52–69.46)	1.00 (reference)			
≥1.5	2226 (14.89)	595 (26.73)	66.18 (65.29–67.08)	1.89 (1.72–2.07)	2.13×10 ^{−40}		
Incident subcohort, N=8953							
Carriers						0.76 (0.75–0.78)	0.001
No	8914 (99.56)	1280 (14.36)	70.54 (69.99–71.09)	1.00 (reference)			
Yes	39 (0.44)	10 (25.64)	63.3 (58.09–68.51)	1.95 (1.04–3.65)	0.04		
GenProb _{CAD} [§]						0.77 (0.76–0.78)	Reference
<0.5	1352 (15.1)	153 (11.32)	73.23 (71.7–74.76)	0.70 (0.59–0.84)	8.44×10 ^{−05}		
0.5–1.5	6251 (69.82)	872 (13.95)	70.89 (70.24–71.54)	1.00 (reference)			
≥1.5	1350 (15.08)	265 (19.63)	67.58 (66.33–68.83)	1.58 (1.37–1.82)	2.73×10 ^{−10}		
Non-European ancestry, N=5526							
Carriers						0.76 (0.75–0.78)	0.63
No	5492 (99.38)	667 (12.14)	64.81 (63.89–65.73)	1.00 (reference)			
Yes	34 (0.62)	6 (17.65)	65 (56.4–73.6)	2.05 (0.91–4.59)	0.08		
PGS _{CAD} [†]						0.77 (0.75–0.79)	0.45
<0.5	202 (3.66)	12 (5.94)	63.33 (56.86–69.8)	0.48 (0.27–0.86)	0.01		
0.5–1.5	4586 (82.99)	533 (11.62)	65.33 (64.3–66.37)	1.00 (reference)			
≥1.5	738 (13.36)	128 (17.34)	62.8 (60.76–64.83)	1.83 (1.50–2.22)	1.67×10 ^{−09}		
PGS _{Lp(a)} [‡]						0.76 (0.74–0.78)	0.50
<0.5	3468 (62.76)	425 (12.25)	65.97 (64.86–67.09)	1.82 (1.61–2.05)	1.03×10 ^{−21}		
0.5–1.5	1128 (20.41)	138 (12.23)	62.83 (60.82–64.83)	1.00 (reference)			
≥1.5	930 (16.83)	110 (11.83)	62.84 (60.34–65.33)	3.43 (2.98–3.95)	3.57×10 ^{−65}		
GenProb _{CAD} [§]						0.78 (0.76–0.79)	Reference
<0.5	507 (9.17)	32 (6.31)	68.88 (65.23–72.52)	0.54 (0.38–0.77)	0.00071959		
0.5–1.5	4036 (73.04)	475 (11.77)	65.31 (64.22–66.4)	1.00 (reference)			
≥1.5	983 (17.79)	166 (16.89)	62.61 (60.78–64.45)	1.68 (1.40–2.01)	1.35×10 ^{−08}		
Incident subcohort, N=3574							
Carriers						0.8 (0.78–0.82)	0.41
No	3552 (99.38)	371 (10.44)	66.02 (64.84–67.2)	1.00 (reference)			
Yes	22 (0.62)	2 (9.09)	64.5 (−43.5 to 172.5)	1.22 (0.30–4.94)	0.78		
GenProb _{CAD} [§]						0.8 (0.78–0.83)	Reference
<0.5	324 (9.07)	18 (5.56)	72.28 (68.1–76.45)	0.64 (0.39–1.03)	0.07		
0.5–1.5	2634 (73.7)	275 (10.44)	66.06 (64.67–67.45)	1.00 (reference)			
≥1.5	616 (17.24)	80 (12.99)	64.42 (61.98–66.87)	1.24 (0.96–1.60)	0.10		

CAD indicates, coronary artery disease; GHI, Genomic Health Initiative; RR, ; HR, hazard ratio; PGS, polygenic risk score; relative risk; and UKB, UK Biobank.

*Adjusted for age, sex, and the first 10 principal components of ancestry.

†Cutoffs are −1.558855 and 1.175333 for 0.5 and 1.5 RR with the total UKB data.

‡Cutoffs are −0.000497 and 0.709967 for 0.5 and 1.5 RR with the total UKB data.

§GenProb at age 75 years with cutoffs of 0.0382527451727196 and 0.130423388954541 for 0.5 and 1.5 RR with the UKB development cohort.

importance of conducting genetic risk assessment early in life. Early identification of elevated genetic risk enables the implementation of personalized prevention strategies at the most effective window before the development of contributing clinical risk factors. Intervention among individuals with higher genetic susceptibility through lifestyle modification and LLT has been consistently shown to reduce the risk of CAD and mitigate associated clinical risk factors.^{2,34–37}

While integrated genetic assessment using monogenic and PRS_{CAD} has been previously reported,^{17,18,38,39} the incorporation of $PRS_{Lp(a)}$ is novel and further enhances the performance of CAD risk stratification by identifying more high-risk individuals. Lp(a) is a well-established causal risk factor for CAD through its proatherogenic, proinflammatory, and prothrombotic effects.^{40,41} Genetic measurement of Lp(a) through $PRS_{Lp(a)}$ is biologically justified, as Lp(a) levels are primarily (>90%) determined by genetic variants in the LPA gene and other loci.^{41,42} Practically, adding $PRS_{Lp(a)}$ to PRS_{CAD} in genetic testing does not increase cost, as both can be calculated from SNPs obtained through the same genotyping or sequencing platforms.

There are several notable limitations in this study. First, we limited the scope of our analysis to genetic risk assessment. This decision was based on the recognition that genetic data and CAD diagnoses are more reliably captured in the biobanks used, whereas information on clinical risk factors and LLT, while important, is often less complete or standardized. Nevertheless, the insights gained from our genetic analyses can inform the design of future studies aimed at integrating genetic and nongenetic risk factors for a more holistic understanding of CAD risk.

Second, despite our efforts to include individuals from minority ancestry groups, the majority of participants in this study were of European ancestry, which may limit the generalizability of our findings. However, the observation of the similar trends across both European and non-European populations within the GHI cohort is encouraging. As larger and more diverse genetic studies of CAD become available, future analyses will be better positioned to refine and extend these findings across a broader range of ancestral backgrounds.

Third, we recognize that a subset of UKB participants was included in the model testing of PRS_{CAD} (PGS003725) and $PRS_{Lp(a)}$ (PGS000667), along with other large cohorts.^{12,15} This may introduce some overfitting bias for these 2 PRSs. However, because the GWAS summary statistics used to develop these PRSs did not include UKB data, the extent of overfitting is likely modest. More importantly, our key findings were independently replicated in the GHI cohort, which was not involved in the development or validation of these PRSs, providing further support for their validity.

Fourth, we acknowledge the subjective nature of selecting RR of 0.5 and 1.5 as cutoffs for defining low

and high risk, respectively, for PRS_{CAD} and $GenProb_{CAD}$. These thresholds were chosen as a proof of principle to demonstrate the effect of categorical PRS on risk stratification. The validity of this cutoff is supported by its similar or higher HR compared with that of monogenic variants.

Finally, it is also acknowledged that individuals who died of CAD before potential recruitment into the biobanks could not be included in the study. This may lead to an underestimation of the risk associated with genetic factors. However, given that the primary aim of the study was to compare the relative effects of various genetic risk factors on CAD risk, and that the age at CAD diagnosis was similar between monogenic variant carriers and individuals with high-risk $GenProb_{CAD}$, the impact of this selection bias on the main findings is likely minimal. Nevertheless, this potential limitation should be considered when interpreting the results. A related methodological consideration is the handling of the competing risk of non-CAD death. In our primary analyses, non-CAD death was treated as a censoring event and cause-specific hazards were estimated; thus, the resulting risk estimates are interpreted conditional on survival. Consequently, absolute CAD risk estimates at older ages may be overestimated. However, the primary objective of this study was comparative genetic risk stratification rather than precise estimation of cumulative incidence, and cause-specific hazards are appropriate for this purpose. Future studies focused on absolute risk prediction may consider alternative competing-risk approaches, such as Fine-Gray subdistribution hazards models.

CONCLUSIONS

We developed and validated a novel integrated genetic risk tool, $GenProb_{CAD}$, that combines monogenic PVs, PRS_{CAD} , and $PRS_{Lp(a)}$ for CAD risk assessment. The substantially improved performance of $GenProb_{CAD}$ compared with monogenic testing alone supports the adoption of an integrated genetic approach to more effectively identify individuals at high genetic risk for CAD in the general population, enabling earlier and more effective prevention strategies.

ARTICLE INFORMATION

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Disclosures

Endeavor Health has agreements with GenomicMD and GoPath Diagnostics for genetic tests of polygenic risk scores. Dr Xu serves a scientific advisory board member for GoPath Diagnostics and GenomicMD. The other authors report no conflicts.

Supplemental Material

Tables S1–S6

Figures S1–S2

REFERENCES

- Zdravkovic S, Wienke A, Pedersen NL, Marenberg ME, Yashin AI, De Faire U. Heritability of death from coronary heart disease: a 36-year follow-up of 20 966 Swedish twins. *J Intern Med*. 2002;252:247–254. doi: 10.1046/j.1365-2796.2002.01029.x
- Khera AV, Emdin CA, Drake I, Natarajan P, Bick AG, Cook NR, Chasman DI, Baber U, Mehran R, Rader DJ, et al. Genetic risk, adherence to a healthy lifestyle, and coronary disease. *N Engl J Med*. 2016;375:2349–2358. doi: 10.1056/NEJMoa1605086
- Jiang X, Holmes C, McVean G. The impact of age on genetic risk for common diseases. *PLoS Genet*. 2021;17:e1009723. doi: 10.1371/journal.pgen.1009723
- Brown MS, Goldstein JL. Familial hypercholesterolemia: defective binding of lipoproteins to cultured fibroblasts associated with impaired regulation of 3-hydroxy-3-methylglutaryl coenzyme A reductase activity. *Proc Natl Acad Sci U S A*. 1974;71:788–792. doi: 10.1073/pnas.71.3.788
- Boren J, Lee I, Zhu W, Arnold K, Taylor S, Innerarity TL. Identification of the low density lipoprotein receptor-binding site in apolipoprotein B100 and the modulation of its binding activity by the carboxyl terminus in familial defective apo-B100. *J Clin Invest*. 1998;101:1084–1093. doi: 10.1172/JCI1847
- Abifadel M, Varret M, Rabes JP, Allard D, Ouguerram K, Devillers M, Cruaud C, Benjannet S, Wickham L, Erlich D, et al. Mutations in PCSK9 cause autosomal dominant hypercholesterolemia. *Nat Genet*. 2003;34:154–156. doi: 10.1038/ng1161
- van der Harst P, Verweij N. Identification of 64 novel genetic loci provides an expanded view on the genetic architecture of coronary artery disease. *Circ Res*. 2018;122:433–443. doi: 10.1161/CIRCRESAHA.117.312086
- Tcheandjieu C, Zhu X, Hilliard AT, Clarke SL, Napolioni V, Ma S, Lee KM, Fang H, Chen F, Lu Y, et al; Regeneron Genetics Center. Large-scale genome-wide association study of coronary artery disease in genetically diverse populations. *Nat Med*. 2022;28:1679–1692. doi: 10.1038/s41591-022-01891-3
- Mack S, Coassin S, Rueedi R, Younsri N, Seppala I, Gieger C, Schonherr S, Forer L, Erhart G, Marques-Vidal P, et al. A genome-wide association meta-analysis on lipoprotein (a) concentrations adjusted for apolipoprotein (a) isoforms. *J Lipid Res*. 2017;58:1834–1844. doi: 10.1194/jlr.M076232
- Sturm AC, Knowles JW, Gidding SS, Ahmad ZS, Ahmed CD, Ballantyne CM, Baum SJ, Bourbon M, Carrie A, Cuchel M, et al; Convened by the Familial Hypercholesterolemia Foundation. Clinical genetic testing for familial hypercholesterolemia: JACC scientific expert panel. *J Am Coll Cardiol*. 2018;72:662–680. doi: 10.1016/j.jacc.2018.05.044
- Khera AV, Won HH, Peloso GM, Lawson KS, Bartz TM, Deng X, van Leeuwen EM, Natarajan P, Emdin CA, Bick AG, et al. Diagnostic yield and clinical utility of sequencing familial hypercholesterolemia genes in patients with severe hypercholesterolemia. *J Am Coll Cardiol*. 2016;67:2578–2589. doi: 10.1016/j.jacc.2016.03.520
- Patel AP, Wang M, Fahed AC, Mason-Suares H, Brockman D, Pelletier R, Amr S, Machini K, Hawley M, Witkowski L, et al. Association of rare pathogenic DNA Variants for familial hypercholesterolemia, hereditary breast and ovarian cancer syndrome, and lynch syndrome with disease risk in adults according to family history. *JAMA Netw Open*. 2020;3:e203959. doi: 10.1001/jamanetworkopen.2020.3959
- Patel AP, Wang M, Ruan Y, Koyama S, Clarke SL, Yang X, Tcheandjieu C, Agrawal S, Fahed AC, Ellinor PT, et al; Genes & Health Research Team; the Million Veteran Program. A multi-ancestry polygenic risk score improves risk prediction for coronary artery disease. *Nat Med*. 2023;29:1793–1803. doi: 10.1038/s41591-023-02429-x
- Misra A, Truong B, Urbat SM, Sui Y, Fahed AC, Smoller JW, Patel AP, Natarajan P. Instability of high polygenic risk classification and mitigation by integrative scoring. *Nat Commun*. 2025;16:1584. doi: 10.1038/s41467-025-56945-0
- Trinder M, Uddin MM, Finneran P, Aragam KG, Natarajan P. Clinical utility of lipoprotein(a) and LPA genetic risk score in risk prediction of incident atherosclerotic cardiovascular disease. *JAMA Cardiol*. 2021;6:287–295. doi: 10.1001/jamacardio.2020.5398
- Dron JS, Wang M, Patel AP, Kartoun U, Ng K, Hegele RA, Khera AV. Genetic predictor to identify individuals with high lipoprotein(a) concentrations. *Circ Genom Precis Med*. 2021;14:e003182. doi: 10.1161/CIRCGEN.120.003182
- Khera AV, Chaffin M, Aragam KG, Haas ME, Roselli C, Choi SH, Natarajan P, Lander ES, Lubitz SA, Ellinor PT, et al. Genome-wide polygenic scores for common diseases identify individuals with risk equivalent to monogenic mutations. *Nat Genet*. 2018;50:1219–1224. doi: 10.1038/s41588-018-0183-z
- Fahed AC, Wang M, Homburger JR, Patel AP, Bick AG, Neben CL, Lai C, Brockman D, Philippakis A, Ellinor PT, et al. Polygenic background modifies penetrance of monogenic variants for tier 1 genomic conditions. *Nat Commun*. 2020;11:3635. doi: 10.1038/s41467-020-17374-3
- Berry ASF, Jones LK, Sijbrands EJ, Gidding SS, Oetjens MT. Subtyping severe hypercholesterolemia by genetic determinant to stratify risk of coronary artery disease. *Arterioscler Thromb Vasc Biol*. 2023;43:2058–2067. doi: 10.1161/ATVBAHA.123.319341
- O'Sullivan JW, Raghavan S, Marquez-Luna C, Luzum JA, Damrauer SM, Ashley EA, O'Donnell CJ, Willer CJ, Natarajan P, American Heart Association Council on G, et al. Polygenic risk scores for cardiovascular disease: a scientific statement from the American Heart Association. *Circulation*. 2022;146:e93–e118. doi: 10.1161/CIR.0000000000001077
- Churchwell K, Elkind MSV, Benjamin RM, Carson AP, Chang EK, Lawrence W, Mills A, Odom TM, Rodriguez CJ, Rodriguez F, et al; American Heart Association. Call to action: structural racism as a fundamental driver of health disparities: a presidential advisory from the American Heart Association. *Circulation*. 2020;142:e454–e468. doi: 10.1161/CIR.0000000000000936
- Bycroft C, Freeman C, Petkova D, Band G, Elliott LT, Sharp K, Motyer A, Vukcevic D, Delaneau O, O'Connell J, et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature*. 2018;562:203–209. doi: 10.1038/s41586-018-0579-z
- David SP, Dunnenberger HM, Choi S, DePersia A, Ilbawi N, Ward C, Wake DT, Khandekar JD, Shannon Y, Hughes K, et al. Personalized medicine in a community health system: the NorthShore experience. *Front Genet*. 2023;14:1308738. doi: 10.3389/fgene.2023.1308738
- Billings LK, Shi Z, Mulford AJ, Wei J, Tran H, Ashworth A, Zheng SL, Dunnenberger HM, Hulick RJ, Sanders AR, et al. Validation of GenProT1D and its clinical utility for differentiating types of diabetes in a biobank from a US healthcare system. *J Diabetes Investig*. 2025;16:10. doi: 10.1111/jdi.14297
- Li JH, Mazur CA, Berisa T, Pickrell JK. Low-pass sequencing increases the power of GWAS and decreases measurement error of polygenic risk scores compared to genotyping arrays. *Genome Res*. 2021;31:529–537. doi: 10.1101/gr.266486.120
- Chora JR, Iacocca MA, Tichy L, Wand H, Kurtz CL, Zimmermann H, Leon A, Williams M, Humphries SE, Hooper AJ, et al; ClinGen Familial Hypercholesterolemia Expert Panel. The Clinical Genome Resource (ClinGen) familial hypercholesterolemia variant curation expert panel consensus guidelines for LDLR variant classification. *Genet Med*. 2022;24:293–306. doi: 10.1016/j.jgm.2021.09.012
- Lambert SA, Gil L, Jupp S, Ritchie SC, Xu Y, Buniello A, McMahon A, Abraham G, Chapman M, Parkinson H, et al. The polygenic score catalog as an open database for reproducibility and systematic evaluation. *Nat Genet*. 2021;53:420–425. doi: 10.1038/s41588-021-00783-5
- Mars N, Lindbohm JV, Della Briotta Parolo P, Widen E, Kaprio J, Palotie A, Ripatti S; FinnGen. Systematic comparison of family history and polygenic risk across 24 common diseases. *Am J Hum Genet*. 2022;109:2152–2162. doi: 10.1016/j.ajhg.2022.10.009

29. Khera AV, Chaffin M, Zekavat SM, Collins RL, Roselli C, Natarajan P, Lichtman JH, D'Onofrio G, Mathera J, Dreyer R, et al. Whole-genome sequencing to characterize monogenic and polygenic contributions in patients hospitalized with early-onset myocardial infarction. *Circulation*. 2019;139:1593–1602. doi: 10.1161/CIRCULATIONAHA.118.035658
30. Taylor A, Wang D, Patel K, Whittall R, Wood G, Farrer M, Neely RD, Fairgrieve S, Nair D, Barbir M, et al. Mutation detection rate and spectrum in familial hypercholesterolaemia patients in the UK pilot cascade project. *Clin Genet*. 2010;77:572–580. doi: 10.1111/j.1399-0004.2009.01356.x
31. Abul-Husn NS, Manickam K, Jones LK, Wright EA, Hartzel DN, Gonzaga-Jauregui C, O'Dushlaine C, Leader JB, Lester Kirchner H, Lindbuchler DM, et al. Genetic identification of familial hypercholesterolemia within a single U.S. health care system. *Science*. 2016;354:aaf7000. doi: 10.1126/science.aaf7000
32. Dikilitas O, Sherafati A, Saadatagah S, Satterfield BA, Kochan DC, Anderson KC, Chung WK, Hebbbring SJ, Salvati ZM, Sharp RR, et al. Familial hypercholesterolemia in the electronic medical records and genomics network: prevalence, penetrance, cardiovascular risk, and outcomes after return of results. *Circ Genom Precis Med*. 2023;16:e003816. doi: 10.1161/CIRCGEN.122.003816
33. Greenland P, Blaha MJ, Budoff MJ, Erbel R, Watson KE. Coronary calcium score and cardiovascular risk. *J Am Coll Cardiol*. 2018;72:434–447. doi: 10.1016/j.jacc.2018.05.027
34. Mega JL, Stitziel NO, Smith JG, Chasman DI, Caulfield M, Devlin JJ, Nordio F, Hyde C, Cannon CP, Sacks F, et al. Genetic risk, coronary heart disease events, and the clinical benefit of statin therapy: an analysis of primary and secondary prevention trials. *Lancet*. 2015;385:2264–2271. doi: 10.1016/S0140-6736(14)61730-X
35. China Kadoorie Biobank Collaborative G. Joint impact of polygenic risk score and lifestyles on early- and late-onset cardiovascular diseases. *Nat Hum Behav*. 2024;8:1810–1818. doi: 10.1038/s41562-024-01923-7
36. Natarajan P, Young R, Stitziel NO, Padmanabhan S, Baber U, Mehran R, Sartori S, Fuster V, Reilly DF, Butterworth A, et al. Polygenic risk score identifies subgroup with higher burden of atherosclerosis and greater relative benefit from statin therapy in the primary prevention setting. *Circulation*. 2017;135:2091–2101. doi: 10.1161/CIRCULATIONAHA.116.024436
37. Damask A, Steg PG, Schwartz GG, Szarek M, Hagstrom E, Badimon L, Chapman MJ, Boileau C, Tsimikas S, Ginsberg HN, et al; Regeneron Genetics Center and the ODYSSEY OUTCOMES Investigators. Patients with high genome-wide polygenic risk scores for coronary artery disease may receive greater clinical benefit from alirocumab treatment in the ODYSSEY OUTCOMES trial. *Circulation*. 2020;141:624–636. doi: 10.1161/CIRCULATIONAHA.119.044434
38. Reeskamp LF, Shim I, Dron JS, Ibrahim S, Tromp TR, Fahed AC, Patel AP, Hutten BA, Stroes ESG, Hovingh GK, et al. Polygenic background modifies risk of coronary artery disease among individuals with heterozygous familial hypercholesterolemia. *JACC Adv*. 2023;2:100662. doi: 10.1016/j.jaccadv.2023.100662
39. Paquette M, Chong M, Theriault S, Dufour R, Pare G, Baass A. Polygenic risk score predicts prevalence of in patients with familial hypercholesterolemia. *J Clin Lipidol*. 2017;11:cardiovascular disease725–cardiovascular disease732 e5. doi: 10.1016/j.jacl.2017.03.019
40. Emerging Risk Factors C, Erqou S, Kaptoge S, Perry PL, Di Angelantonio E, Thompson A, White IR, Marcovina SM, Collins R, Thompson SG, et al. Lipoprotein(a) concentration and the risk of coronary heart disease, stroke, and nonvascular mortality. *JAMA*. 2009;302:412–423. doi: 10.1001/jama.2009.1063
41. Clarke R, Peden JF, Hopewell JC, Kyriakou T, Goel A, Heath SC, Parish S, Barlera S, Franzosi MG, Rust S, et al; PROCARDIS Consortium. Genetic variants associated with Lp(a) lipoprotein level and coronary disease. *N Engl J Med*. 2009;361:2518–2528. doi: 10.1056/NEJMoa0902604
42. Tsimikas S. A test in context: lipoprotein(a): diagnosis, prognosis, controversies, and emerging therapies. *J Am Coll Cardiol*. 2017;69:692–711. doi: 10.1016/j.jacc.2016.11.042



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